

LEARNING OTCOMES!

Candidates should be able to:

- identify the main locations of limestone, gypsum and rock salt extraction from a map, and understand their uses
- identify the main metallic and non-metallic mineral resources of Pakistan, and in what quantities they: - are extracted - exist as reserves
- understand the extent to which these can be exploited
- describe the environmental problems caused by mineral extraction
- evaluate the benefits of developing mineral resources and understand the sustainability of extraction
- identify the main imported minerals, where they come from, and in what quantities.

MINERALS

- ✓ A mineral is defined as being a naturally occurring element or compound that is formed by inorganic processes and contains a crystalline structure.
- ✓ Minerals have following features;
 - They are solids.
 - They are formed naturally in the earth.
 - They are not alive or made of living things.
 - They have definite shape.

HOW MINERALS ARE FORMED?

- ✓ Either by solidification of molten rocks i.e. from magma and lava. When magma solidifies in the surface it forms Plutonic or Intrusive rocks. (They are both types of rigorous rocks). They can be changed into different forms by heating and compression, and are converted into metamorphic rocks. When lava solidifies it forms Volcanic rock and Exclusive rocks.
- ✓ Some are formed by evaporation of water. When water evaporates it leaves behind crystals of salts which forms rock salt. Rock salt is a sedimentary rock.
- ✓ Some are formed by heating and compression of rocks. When limestone compresses it forms marble. Limestone is also known as calcium carbonate.
- ✓ When there is slow heating of rocks in the subsurface they form large crystals and form gemstones.

MINERAL EXPLORATION

- ✓ To discover the metallic or non metallic minerals from the earth.

MINERAL EXTRACTION

- ✓ Taking out the minerals from the earth by different methods.

MINERAL EXPLOITATION

- ✓ Means to utilize the available mineral resources for development purposes.

LIMESTONE

Uses

- ✓ Cement
- ✓ To neutralize acidic soil conditions
- ✓ As a reagent in flue gas desulfurization (sulphur dioxide air pollution control)
- ✓ Added to paper, plastics, paint, tiles
- ✓ Toothpaste
- ✓ Used in blast furnaces to extract iron from its ore
- ✓ Medicines
- ✓ Cosmetics



Areas

- ✓ Daud khel, Zinda pir, Kot Digi, Harnai, Ganjotaker, Mangopir and Murlihills.

GYPSUM

Uses

- ✓ Cement
- ✓ Gypsum Board primarily used as a finish for walls and ceilings
- ✓ Fertilizer and soil conditioner
- ✓ Plaster of Paris (surgical splints; casting moulds; modelling)
- ✓ In foot creams, shampoos and many other hair products.



Areas

- ✓ It is mined from Khewra, Dandot and Daudkhel.

ROCKSALT

Uses

- ✓ Cooking
- ✓ Manufacturing pulp and paper
- ✓ Producing soaps, detergents, and other bath products
- ✓ To preserve fish after it has been caught



Areas

- ✓ Important mines are Khewra, Warcha and Kalabagh

BRINE

- ✓ Mixture of Rocksalt and Water

LOCATION OF LIMESTONE, GYPSUM, ROCKSALT

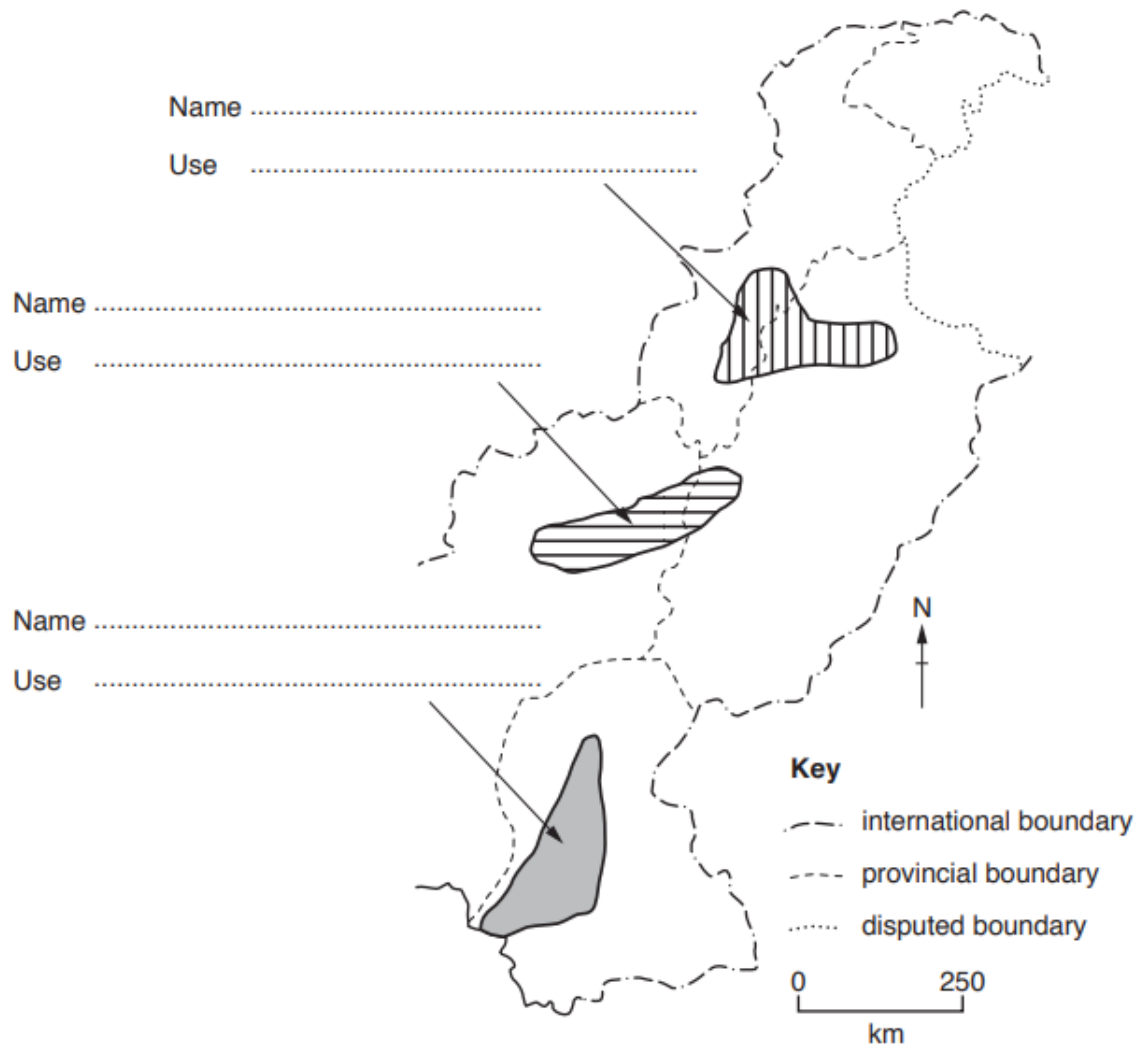


Fig. 3

- ✓ Limestone available at all 3 places
- ✓ Gypsum only available at 2 places; Southern and Central.
- ✓ Rocksalt only available at 1 place; Northern.

Difficulties there may be in getting minerals to export markets.

- ✓ Heavy / bulky commodities
- ✓ Expensive to transport
- ✓ Roads and railways from mining areas poorly developed
- ✓ Mostly extracted far inland, away from ports
- ✓ Mountainous / rugged terrain
- ✓ Theft

- ✓ Inappropriate / inadequate vehicles to transport minerals

METALLIC MINERALS

- ✓ Metallic minerals are those which are made up of a metallic compound e.g. Iron, copper.
- ✓ Economically more valuable e.g. Gold.
- ✓ Generally hard, tough and shiny e.g. Iron.
- ✓ Can change shape without breaking e.g. copper
- ✓ Can be stretched and compressed e.g. Iron.
- ✓ Many are good thermal and electrical conductors e.g. copper.
- ✓ More reactive with water and acid e.g. copper.

NAME	DEPOSITS	QUALITY	EXTRACTION
Aluminium	74 million tonnes	Low to medium	2,000 tonnes
Iron Ore	900 million tonnes	Low to high	24,000 tonnes
Copper	1900 million tonnes	Low to high	15,000 tonnes
Chromite	2.53 million tonnes	Low to medium	27,000 tonnes
Gold	11.2 million ounces	Low Grade	-

CHROMITE

Uses

- ✓ It gives hardness and electrical resistance to steel.
- ✓ It is used for bridges and railway carriages.
- ✓ It is used as lining in metallurgical furnaces and for making engineering tools and stainless steel.

Area

- ✓ Muslim Bagh

Suggest why the production of minerals, such as chromite, varies from year to year.

- ✓ Investment / funding varies
- ✓ Demand / orders vary
- ✓ Bankruptcy / companies leave
- ✓ Problems with machinery
- ✓ Reserves reducing / new reserves exploited

- ✓ Terrorism

Jab bhi vary ka sawal aayega yehi likhna hai for any mineral.

IRON ORE

- ✓ It is used in steel making, construction and in transport industry.

COPPER

- ✓ It is used to make copper wire and other electrical appliances specially switches, that carry current.
- ✓ It is also used in making Alloys, water pipes and tanks.

MANGANESE

- ✓ It is used in making dry cell and paints.

BAUXITE

- ✓ Aluminium is mainly obtained from bauxite and is a valuable metal.
- ✓ It is used in utensils, tins, cans and many other products.

CELESTITE

- ✓ It is used in tracer bullets, fireworks, ceramics, paints and plastics.

NON-METALLIC MINERALS

- ✓ Non-Metallic minerals are those which are made up of a non metallic compound e.g. Limestone, Gypsum.
- ✓ Economically less valuable except power resources e.g. oil, gas and coal
- ✓ Softer, rough and may not shine e.g. clay.
- ✓ Breaks away when shape is changed except oil and gas.
- ✓ Cannot be stretched or compressed except natural gas
- ✓ Poor thermal and electrical conductors e.g. rock salt.
- ✓ Less reactive with water and acid except limestone and salt

NAME	DEPOSITS	QUALITY	EXTRACTION
Limestone	Large	High	8,697,573 tonnes
Gypsum	4850 million tonnes	Medium to high	384,513 tonnes
Rocksalt	600 million tonnes	High	1 million tonnes
Coal	185000 million tonnes	Low to medium	3 million tonnes
Oil	300 million barrels	Good	23 million barrels
Gas	27 trillion cubic feet	Good	900000 million cubic feet

MARBLE

- ✓ It is found in bands of white, grey, yellow and brown.
- ✓ It is used in buildings and for making chips for flooring and decorative pieces.

SULPHUR

- ✓ It is used to manufacture sulphuric acid, explosive materials, paints, dyes, pulp, fertilizers and is used in the refining of petroleum.

CLAYS

- ✓ It can be differentiated into China Clay, Fire Clay and Fuller's Earth.
- ✓ China Clay is used in ceramic industry.
- ✓ Fire clay is used for insulation and preparation of bricks and also used in making pottery.
- ✓ Fuller's Earth is used in steel mills, oil drilling and oil refining.

SOAPSTONE

- ✓ It is found in the form of stone and then crushed to make powder.
- ✓ It is used as a filler in soap, ceramics and face powder

BARITES

- ✓ It is used in oil industry and for manufacturing the paints, glass and insecticides.

EFFECTS OF MINING ON ENVIRONMENT

- ✓ Deforestation.
- ✓ Vegetation cut down resulting in soil exposure
- ✓ Natural landscape deformed due to construction of roads and miners' houses
- ✓ Rock blasting or digging of earth.
- ✓ Depressions caused by subsidence of land may become flooded.
- ✓ Noise pollution and ground vibration from blasting.
- ✓ Traditional mining methods are hazardous to the health of miners.
- ✓ Land pollution due to mining waste.
- ✓ Water supply polluted from mineral waste.
- ✓ Air pollution from dust and smoke.

SOLUTION OF THESE PROBLEMS

- ✓ Mining should be done on scientific lines to maximize the benefits and minimize the negative effects.
- ✓ When mining is done in an area, special precautions must be taken to prevent environmental losses.
- ✓ The land should be leveled and the depressions should be filled in after the mining activity to avoid deformation.
- ✓ All the mining waste including gases, fumes and semi-solid waste should be properly treated before their final disposal.
- ✓ The people involved should be provided with proper protective clothing to prevent the effects of gases and dust.
- ✓ The area around the mine should be properly planted with trees so that the atmosphere may remain healthy and when miners come out of the mines, they may have fresh air to breathe
- ✓ Water treatment plants should be set up to supply clean water to the workers living near mines.

BENEFITS OF MINERALS

- ✓ Increase GNP (Gross National Product)
- ✓ It encourages foreign investment.
- ✓ Provides raw materials for industries.
- ✓ Decrease imports.
- ✓ It saves and earns foreign exchange.
- ✓ Provides power for industries.
- ✓ Provides fuel for transport and homes.
- ✓ Provides employment opportunity.
- ✓ It reduces rural urban migration.
- ✓ It encourages the development of areas where they are formed.
- ✓ Reduces National debts.
- ✓ Revenue/Taxes earned by Government.

WHY PAKISTAN IS BACKWARD IN MINERAL?

HUMAN FACTORS

- ✓ Infrastructure is very poor in Pakistan. Many areas do not have roads and railways, especially those in Northern areas and Baluchistan.
- ✓ There is also lack of electricity and water supply.
- ✓ Lack of skilled labor in Pakistan.
- ✓ Finance shortages
- ✓ Poor planning
- ✓ Heavy reliance on Multi-national companies which takes most of the profit.
- ✓ Inadequate planning and geology mapping.
- ✓ Institutional Mismanagement and corruption.
- ✓ Frequent change of government in previous years which do not have consistent policies related to mineral sector.
- ✓ Low priority given to mineral sector.
- ✓ Lack of security and terrorism which discourages investors.
- ✓ Many areas of Baluchistan and FATA are not under central Administration and the tribal chiefs do not want any development in their area. Hence, in the long run people can stand against them.

NATURAL FACTORS

- ✓ Many areas have rugged topography. They are cut off by mountains, deserts and glaciers such as Baluchistan and Northern Areas.
- ✓ Many areas have extreme climatic conditions. They are either very hot or very cold. Hence, exploration can't be carried out. Labor are not ready to work in extreme conditions.
- ✓ Geology is very complex. There is much folding and faulting.

To what extent can more extraction of mineral resources help to increase development in Pakistan? [6]

- ✓ In Possibilities (Benefits)
- ✓ In Difficulties (Why Pakistan Backward?)

SUSTAINABLE MINING

- ✓ The available mineral resources meet the needs of the present and the future generation but not harm the environment.
- ✓ Mining should be done on modern lines to maximize the benefits and minimize the negative effects.
- ✓ The discharge of toxic substances and the release of heat, which is harmful to environment should be strictly checked.
- ✓ Proper training should be given to workers.
- ✓ The land should be leveled and the depressions should be filled in after the mining activity to avoid deformation.
- ✓ Hi-tech knowledge and experts should be available.
- ✓ Provision of infrastructure (transport, power, telephone lines, water supply etc).
- ✓ Explore more mineral resources.
- ✓ Reprocess the mineral waste.

UNSUSTAINABLE MINING

- ✓ The available mineral resources could not meet the needs of the present and the future generation but harm the environment.
- ✓ Deforestation
- ✓ Pollution i.e air, water and noise

- ✓ Soil erosion
- ✓ Lack of infrastructure.
- ✓ Corruption

MAIN IMPORTED MINERALS

IRON AND STEEL

- Pakistan imports around \$400 million worth of iron and steel related products annually from countries like Japan, Russia, and Ukraine etc

COAL

- Pakistan imports around \$300 million of coal and related products from countries like South Africa and Indonesia

CRUDE OIL AND PETROLEUM PRODUCTS

- Pakistan imports around \$5 billion worth of these products, from countries like Saudi Arabia, UAE and Qatar etc